

CivicPulse

Smart Governance Platform

MY ROLE

Full-Stack Development

OBJECTIVE

Develop a platform that enables citizens to report and manage civic issues efficiently.

KEY AREAS OF CONTRIBUTION



User
Authentication



Backend API
Development



Core CivicPulse
Features



JWT & Spring
Security

APPLICATION PREVIEW



The live CivicPulse home screen — the entry point where citizens sign in and report issues.



WORK COMPLETED

- Designed and developed the user registration functionality
- Implemented user login
- Connected the React frontend with the Spring Boot backend
- Implemented API communication between client and server
- Used MySQL for storing user information
- Configured React Router for navigation

KEY APIS

`/api/auth/register``/api/auth/login``/api/users`

ACTUAL IMPLEMENTATION

Register

Login



Developed the backend foundation of CivicPulse using Java Spring Boot.

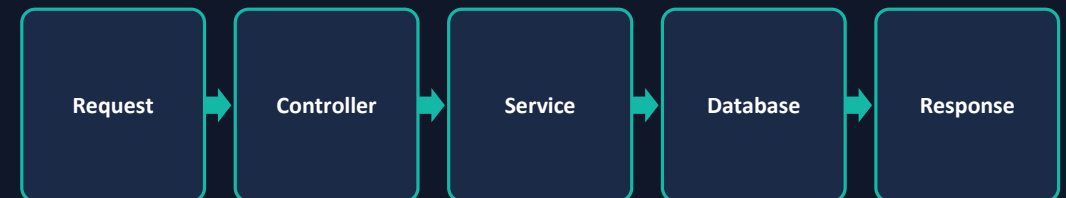
WORK COMPLETED

- Created REST APIs for the application
- Implemented backend controllers and services
- Connected the application with MySQL
- Implemented database operations using Spring Boot
- Tested and verified APIs using Postman
- Integrated backend APIs with the React frontend

LAYERED ARCHITECTURE



REQUEST LIFECYCLE





CORE FEATURES IMPLEMENTED

- Citizen issue reporting
- Viewing reported civic issues
- Managing issue information
- Citizen / admin interaction with civic issue data
- Backend support for core CivicPulse operations
- Integration between frontend, backend, and database

ACTUAL IMPLEMENTATION

Report Civic Issue

Reported Issues

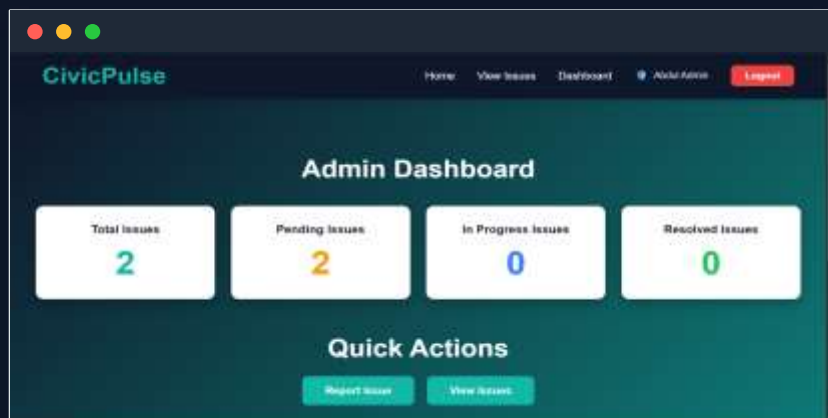
Focused on end-to-end application functionality — from citizen reporting to issue tracking.



SECURITY IMPLEMENTATION

- Integrated Spring Security with the Spring Boot backend
- Implemented JWT-based authentication
- Backend verifies login credentials
- A JWT token is generated after successful authentication
- Token is stored on the client side
- JWT is sent with protected API requests
- Backend validates the JWT before allowing access to protected resources

PROTECTED RESOURCE — ADMIN DASHBOARD



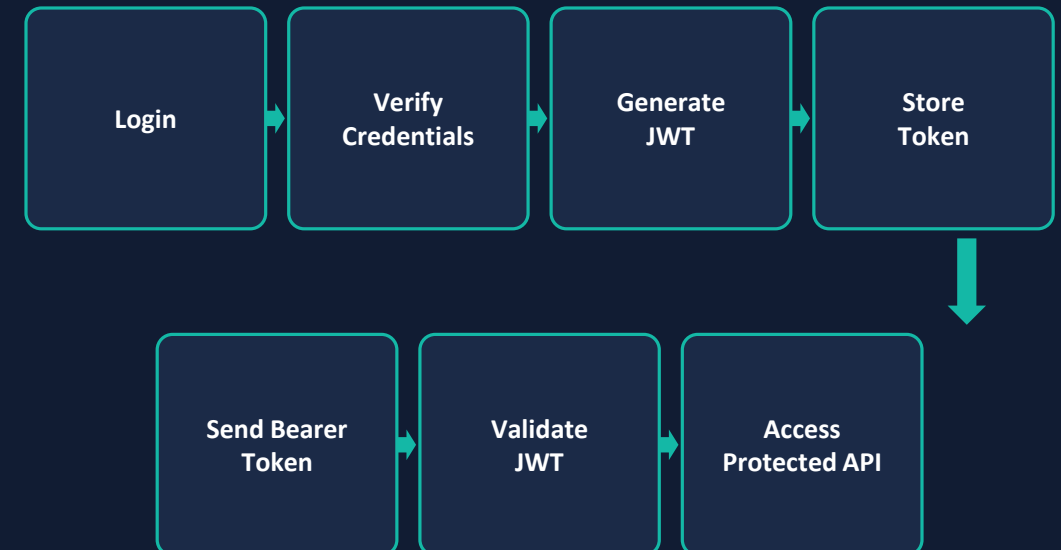
Tested in Postman



Protected Resources

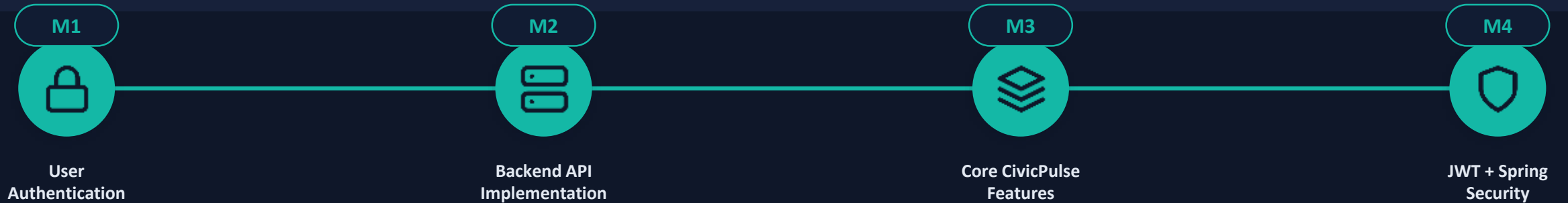
Authorization: **Bearer** <JWT Token>

AUTHENTICATION & AUTHORIZATION FLOW



Internship Journey & Key Learnings

CivicPulse — Smart Governance Platform



MAJOR TECHNICAL LEARNINGS

- ✓ React.js & frontend-backend integration
- ✓ MySQL database integration
- ✓ Spring Security
- ✓ Java Spring Boot
- ✓ API testing with Postman
- ✓ Full-stack application development
- ✓ REST API development
- ✓ JWT authentication
- ✓ Debugging & API integration



Outcome: Successfully contributed to the development of CivicPulse — from authentication and backend APIs to core civic features and secure JWT-based access.